SAMSUNG PRISM GENAI HACKATHON · THEME 05 · ROUND 2

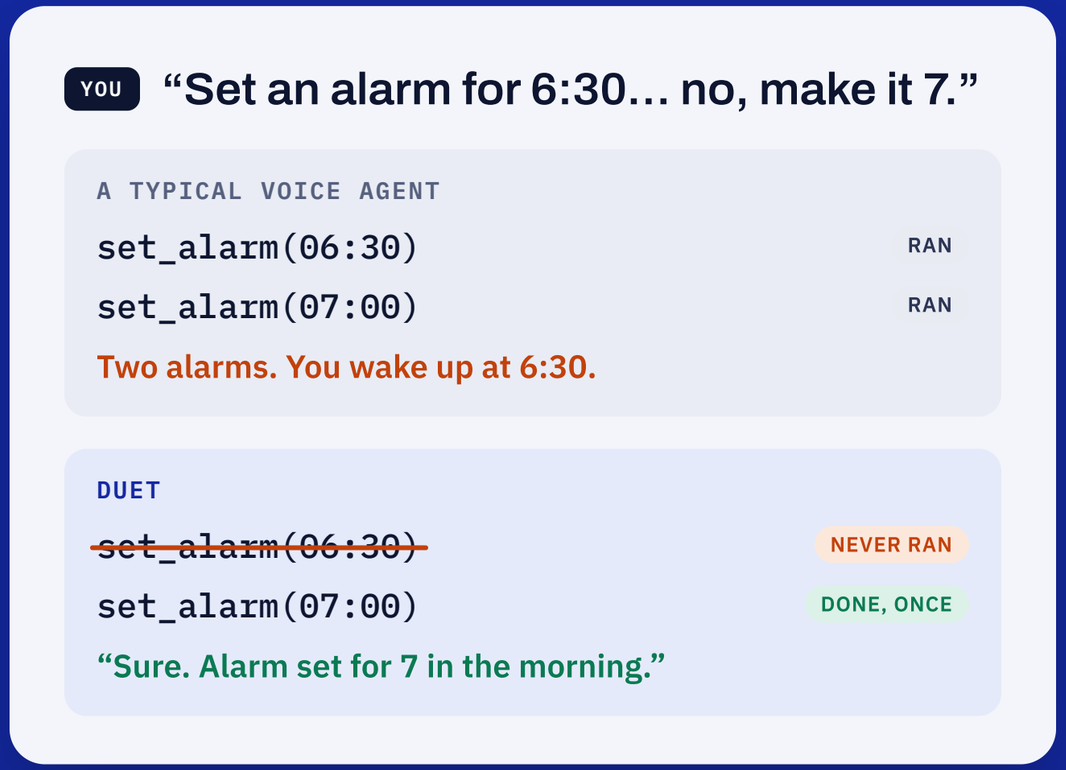
DUET

Interrupt it. Hesitate. Change your mind.

A full-duplex voice agent for Galaxy phones that acts on what you finally meant: once, and only after the phone confirms it.

Team SE7EN · SRM Institute of Science and Technology

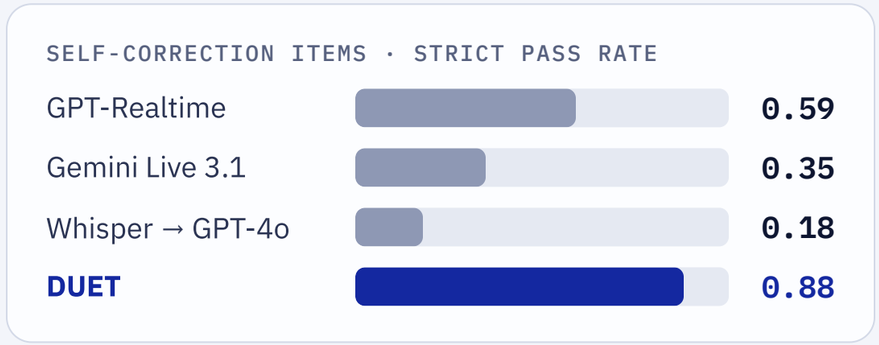
Tanmay Singh · Panshul Arora · Naman Rai · Devansh Goenka



Voice agents act before the user has finished deciding

> **“Models commit intermediate parameters before the correction arrives.”**

Full-Duplex-Bench v3 (NTU and NVIDIA, 2026), on why today's voice agents fail at tool use.



SIX CAVEATS WE FOUND WHILE BUILDING IT, AND WHAT ANSWERS EACH

A correction arrives after the plan

“6:30… no, make it 7”: the first value is already on its way.

`epochs · commit gate`

Not every sound is a turn

A cough, room noise, a listener's “mm-hm”.

`only words void a plan`

Cut off mid-answer

What did the user actually hear before they spoke?

`heard-position tracking`

A retry can do it twice

The text to Gupta Hostel times out. Did it go? Sending it again may send it twice.

`exactly-once ledger`

Fast and smart pull apart

“Torch on” needs an answer in 0.3 s; “Why is my battery draining?” needs real reasoning.

`two minds · code checks each plan`

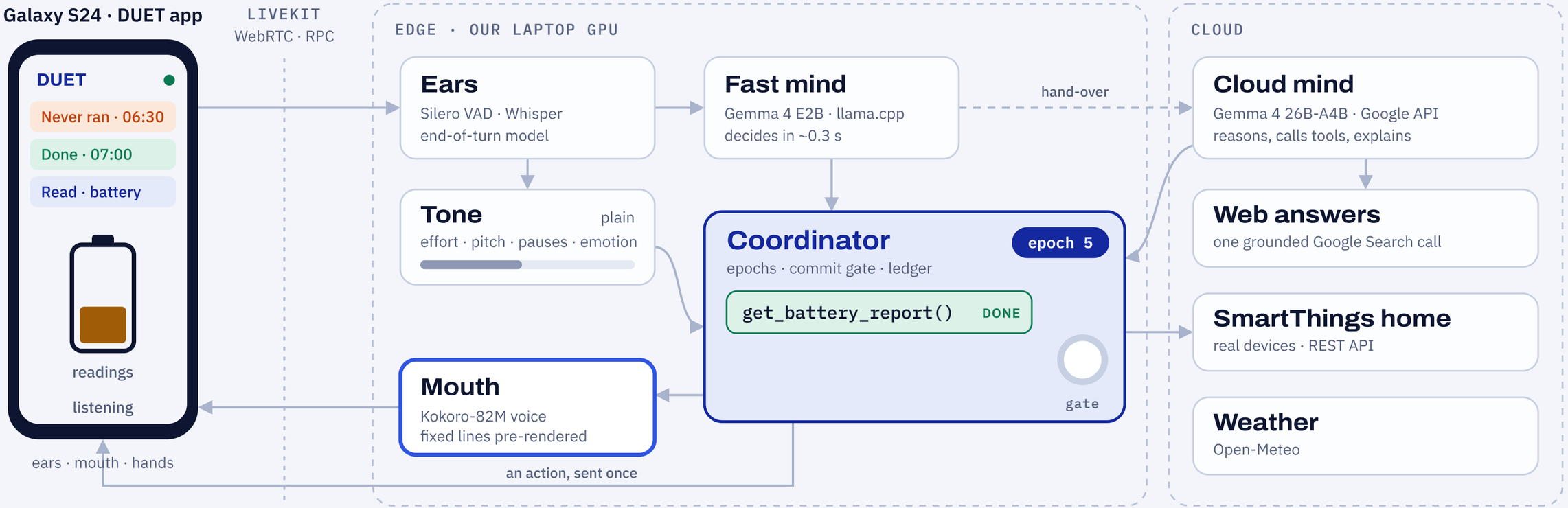
The words hide the doubt

“Set it for 4:30… I think?” The words say 4:30; the voice says not sure.

`tone channel · asks first`

The tension: reply within about a second, but act only on what the user finally means.

One conversation through DUET



YOU “Mm-hm.”

DUET “Mostly, yes. Turning the brightness down would help the most, `mm-hm` and closing that game would help too.”

7 · A BACKCHANNEL IS NOT AN INTERRUPTION

“Mm-hm” lets the answer **carry on** from the sentence it was in. “Wait”, “go on” and “what?” are understood **by code, with no model call**.

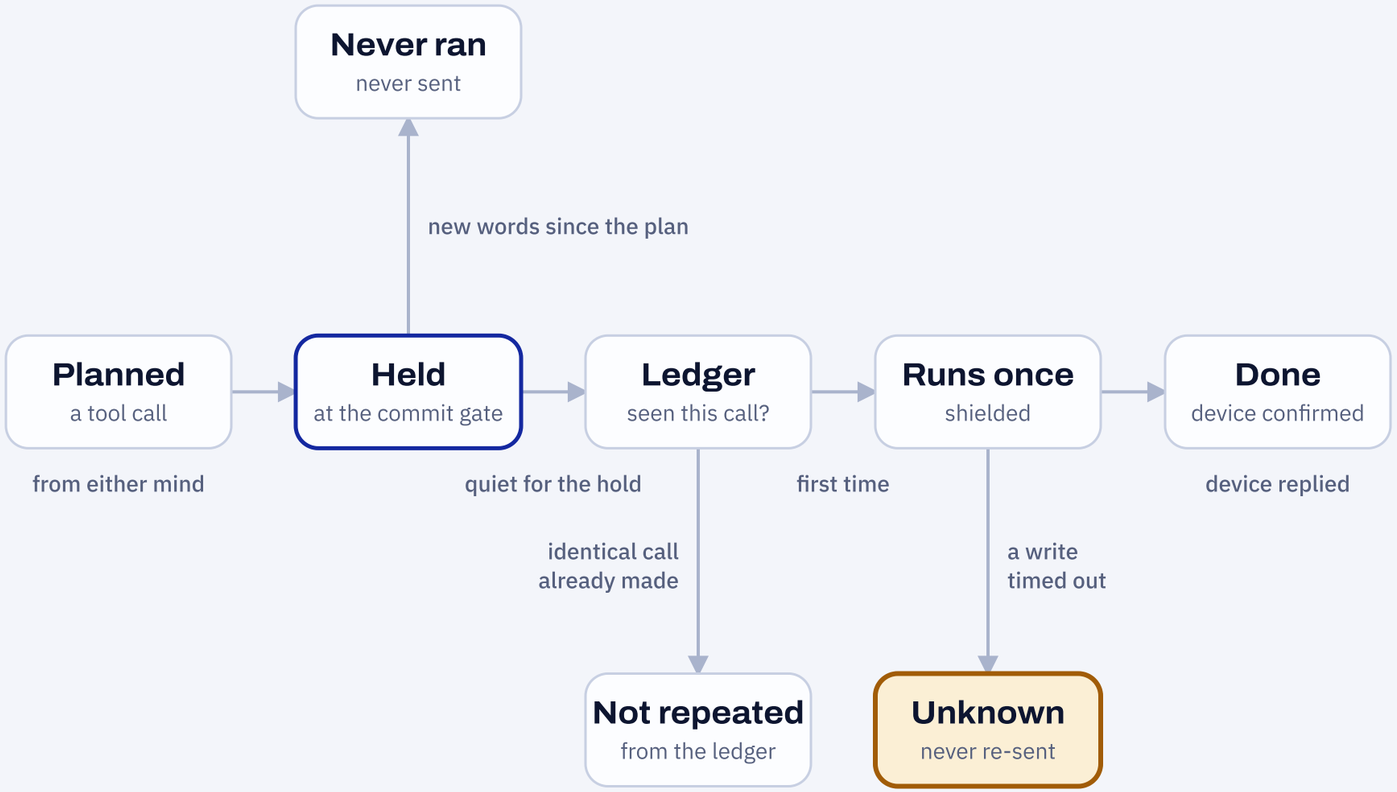
AFTER YOU STOP SPEAKING

0 calls

```
“mm-hm”  → carries on
“wait”   → stops, stays quiet
“go on”  → finishes the answer
“what?”  → says it again
```

The coordinator: the model proposes, code decides

ONE PROPOSED ACTION



Epochs

Every plan carries the epoch of the words it was made on. New words void it, even after the newer turn has closed. A cough brings no words, so it only delays.

Commit gate

No action while you speak. Thinking starts at once; acting waits until you have been quiet for:

`1.4 s` `1.8 s after “no wait”, “make it”` `2.2 s after “and…”`

Idempotency ledger

The same call is answered from the ledger, never sent twice. A call past the gate finishes and is recorded even if you barge in.

Failure policy

A failed read is retried once. A failed or timed-out write is never re-sent: DUET says it may not have gone through.

It hears what you say and how you say it



Tone may make DUET wait or ask. It never runs or cancels an action on its own: every action still needs words.

Unsure — BUILT

“Set an alarm for 4:30, I think.”

Asks “4:30 in the afternoon, right?” before acting. “Yes” then sets it, with no model call.

`words: a hedge next to the value · voice: a long pause before it`

Urgent — BUILT

“No, no, no!” shouted over an answer

Stops at once and stays quiet. Pending actions wait for the words before anything runs.

`voice: brighter +13 dB, higher +9 semitones · words: “no”`

Distressed — BUILT

“My knee hurts…” in a low, shaky voice

Answers gently, asks how they are, offers to call family, and speaks 10% slower.

`voice: emotion model · words: pain`

Mixed signals — NEXT

“I'm fine” said shakily

The words and the voice disagree: check in gently and offer to alert family.

`the voice must outweigh the words, so it needs care`

Sarcasm — NEXT

“Oh great, set it for 5 AM.”

At most a quick yes/no check before acting. Never acted on.

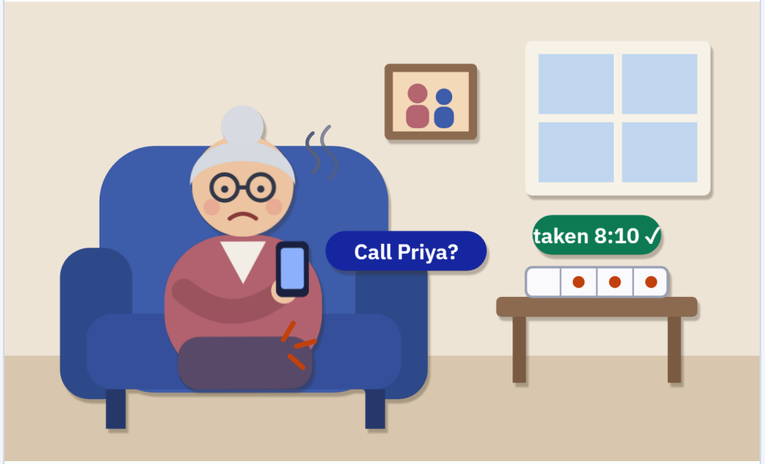
`a research-grade cue, so advisory only`

Why brightness and pitch, not loudness: a phone's gain control flattens loudness. On CREMA-D actors, angry speech was only +2.3 dB louder after levelling, but +18.9 dB brighter and +14 semitones higher. Each segment is measured in about 12 ms, in parallel with speech recognition.

On a Galaxy today: real phone actions, a real SmartThings home

Care

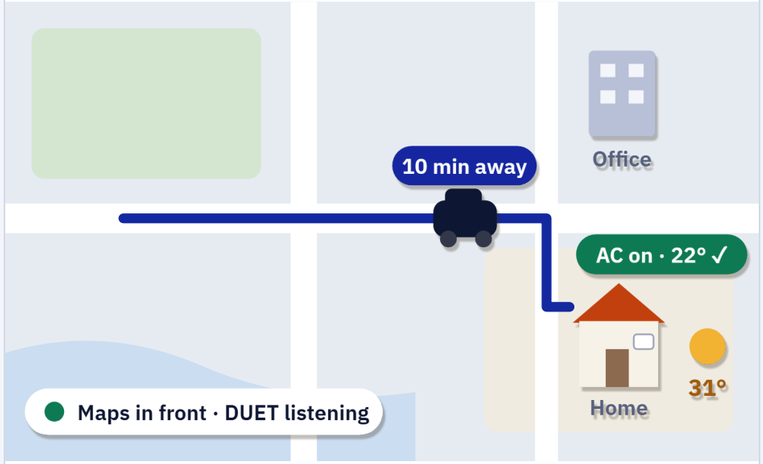
an older person living alone



ASHA “My knee hurts today…” *(low, shaky)*
DUET “I'm sorry it hurts. Is it very painful? Shall I call Priya?”

Drive

eyes on the road



YOU “Yes, make it 22.”
DUET “The living room AC is on at 22°.” Confirmed by SmartThings.

Assistant

every Galaxy owner



YOU “Okay, bring the brightness down to 40.”
DUET “Brightness set to 40%.”

BUILT ON PUBLIC ANDROID AND SAMSUNG INTERFACES

DUET's own alarms, on the lock screen · timers · calendar · calls and texts · Google Maps, listening behind it · brightness · timeout · text size · Do Not Disturb · torch · media · volume · battery and app diagnostics · **SmartThings devices** · **offers when you are nearly home** · weather · web answers · **the side-key assistant**

From an edge prototype to the safety layer inside Samsung's agents

TODAY

BUILT · IN THE DEMO

- **Galaxy app:** the ears, mouth and hands, on the side key
- **Edge:** speech, the fast mind, the coordinator, tone
- **Cloud:** Gemma 4 26B, web answers, SmartThings
- **Three modes**, skills from a manifest

PHASE 1 · MONTHS 0–3

All of it on the phone

- The coordinator in **Kotlin**, with the same tests
- **Gemma 4 E2B on the phone** through Google's LiteRT-LM (52 tokens/s published on the Galaxy S26 Ultra)
- Streaming speech recognition and the tone channel on the phone

DONE WHEN
a simple action lands in under 1.5 s with the laptop switched off, and the spoken suite still shows 0 stale and 0 duplicate actions

PHASE 2 · MONTHS 3–6

Listening like a person

- **Hindi and Hinglish**, in recognition and in the fast mind
- **Gemma hears the audio itself:** meaning and tone from one model
- Mixed signals and sarcasm, as cues that can only ask
- Your voice's baseline kept on the phone, never uploaded

DONE WHEN
Hinglish corrections pass at the English rate, and tone asks back on at most 1 plain request in 20

PHASE 3 · MONTHS 6–12 · WITH SAMSUNG

Inside Samsung's agents

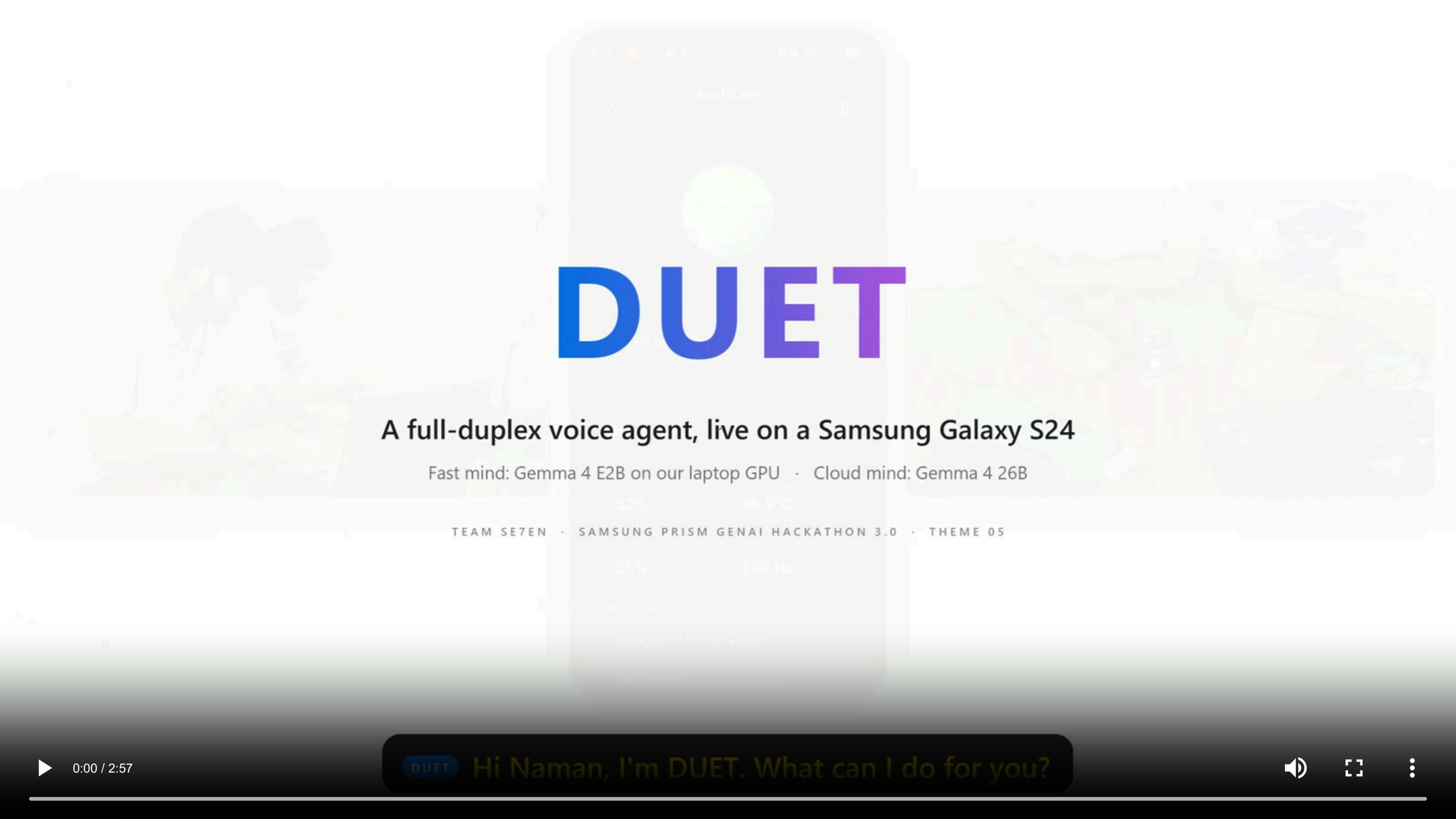
- **Bixby's device agents and AppFunctions:** the coordinator in front of every action
- **SmartThings Car-to-Home:** offers that start in the car, settled by your words
- **Galaxy Watch and Buds:** Care check-ins and family alerts
- **Samsung SDS voicebots:** book once, never twice

DONE WHEN
every action an agent takes passes the gate, the ledger and the device's own confirmation

Why it scales: a new capability is one entry in a manifest, and it inherits the gate, the ledger and the device's confirmation, with no new code.

Models are configuration.
The coordinator is the product.

The live demo →


DUET
A full-duplex voice agent, live on a Samsung Galaxy S24
Fast mind: Gemma 4 E2B on our laptop GPU · Cloud mind: Gemma 4 26B
TEAM SE7EN · SAMSUNG PRISM GENAI HACKATHON 3.0 · THEME 05
0:00 / 2:57
DUET
Hi Naman, I'm DUET. What can I do for you?

APPENDIX · FOR QUESTIONS

The detail behind the design

A1 Where the time goes, and why the AI runs on the edge today

A2 How the fast mind is kept safe

A3 Tone: what we measured before building it

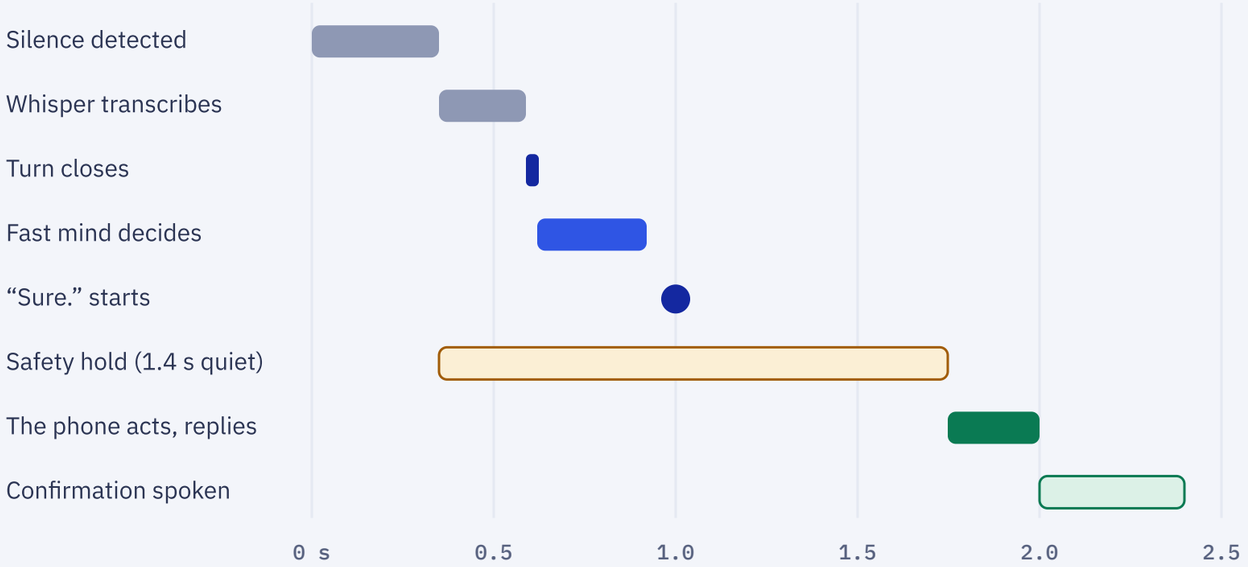
A4 Tested like a product: suites and failure drills

A5 Full-Duplex-Bench v3 in detail

A6 Where DUET sits among voice agents

Where the time goes, and why the AI runs on the edge today

A simple action, seconds after your last word



Medians from our runs. Thinking and the hold overlap: the hold is counted from the end of speech, so safety costs little time. A correction (“no, make it 7”) raises the hold to 1.8 s; a cloud answer adds one Gemma 4 26B call, 1.5–2 s at the median.

Each module placed by measurement

MODULE	ON THE EDGE TODAY	ON THE S24 ITSELF
Fast mind, Gemma 4 E2B	`0.24 s a decision`	≈3.3 s at published S24-class rates
Speech recognition	Whisper large-v3-turbo	streaming models ≤0.3 s faster, much less accurate on Indian English
Voice	Kokoro, streamed	no gain from Android's voice
Phone ↔ edge	`90 ms round trip`	an action and its reply: 253 ms
Cloud mind	`1.5–2 s a call`	the same call, from the phone

The AI moves onto the phone once phones run Gemma fast enough. Gemma 4 E2B runs at 52 tokens/s on the Galaxy S26 Ultra (published), and the coordinator was written to port.

How a 2-billion-parameter model is kept safe

What the fast mind may do

- **Answer small talk itself** (“Good morning”, “I'm tired”), after a code check: no number, no health advice, no claim, no tool name. 0.25–0.45 s.
- **Act** on a simple phone action: alarm, timer, torch, brightness, a call, a text, Maps, a home device.
- **Keep listening** when the words stop mid-phrase (“Call my…”).
- **Hand over** with a bridge line in its own words and a kind: **web** → one grounded search · **health or talk** → one light call · **phone** → the full cloud mind with tools.
- **Plain readings** (“How much battery do I have?”) are answered from the reading itself, with no model call.

Code checks every plan before it runs

- **Named in your words:** “how many shopping apps do I have?” is not open_app(apps).
- **Values you said:** a contact, place or number you never said goes to the cloud mind.
- **The last value said is in the plan:** “3:30, no, cancel that, 4:30 instead” must plan 4:30.
- **A question is answered, never acted on:** “Is the AC on?”
- **Nothing dropped, nothing earlier changed:** “…actually never mind” and “make the alarm 7” go to the cloud mind, which has the history.

Measured: **188 of 190 decisions right, 0 wrong, 2 handed over;** actions decided in 0.18–0.24 s. A plan that fails a check costs a second, never a wrong action.

What we measured before building the tone channel

Vocal effort that survives a phone's gain control

CREMA-D ACTORS, LEVELLED	BRIGHTNESS	PITCH	LOUDNESS	MARKED RAISED
Angry	+18.9 dB	+14.0 st	+2.3 dB	9 of 10
Fearful	+13.3 dB	+12.3 st	+2.5 dB	7 of 10
Happy	+12.2 dB	+9.8 st	+1.3 dB	8 of 10
Sad	−1.0 dB	+1.4 st	−1.0 dB	0 of 10
Neutral	0	0	0	0 of 10

Each take against the same actor's neutral one. Plain speech marked raised: 0 of 13 of our own S24 utterances, 0 of 40 synthetic lines. Excitement sounds like anger, so a raised voice alone changes nothing but the timeline.

Models and cues we tried

- **SenseVoice-Small:** “unknown” emotion on 49 of 50 English clips. Rejected.
- **emotion2vec, distilled to 9.7 MB:** 19 of 30 negative clips caught, 1 false alarm in 20. Used only together with distress words.
- **A rising pitch at the end:** flagged 6 of 29 plain statements; Indian English often ends rising. Off by default.
- **Our own pitch tracker** made octave errors, so we use Praat.
- **Cost:** 12 ms a segment at the median, 31 ms at most, in parallel with Whisper.

Tested like a product, including failures on purpose

The layers

- **345 unit tests:** the gate, epochs, ledger, guards, routing, key pool and tone.
- **66 spoken scenarios:** a synthetic voice talks to the full live stack; each scenario says what DUET must do, must never do and must never claim.
- **46 of 46 end-to-end cases** on Android 16 with the real app, also behind Google Maps.
- **190 fast-mind decisions,** with held-out sentences written after its checks.
- **One command** runs all of it and writes one report.

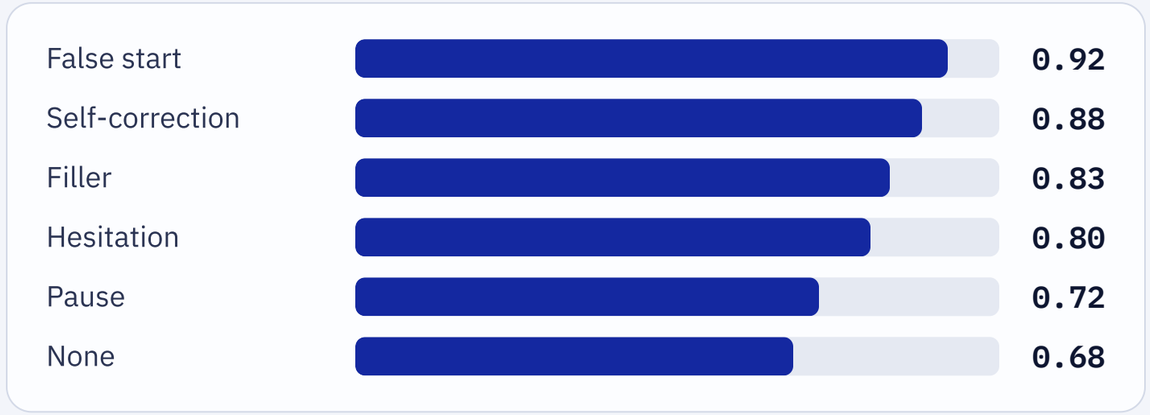
Across 200+ spoken test conversations: 0 stale and 0 duplicate actions.

Switched off on purpose

WHAT FAILED	WHAT DUET DID
The cloud mind	Said so at once; simple actions kept working on the fast mind
The fast mind's server	Every request went to the cloud mind, with no wait
One of two API keys	Every request answered on the good key; the bad one rests
The phone refuses	“I couldn't turn the torch on, but you can allow this in settings.”
The phone never answers	“It may not have gone through”: never claimed, never re-sent
The room drops behind Maps	Rejoined in 3.7 s, still listening

The benchmark in detail

Pass@1 by kind of disfluency



By tools expected: 1 tool 0.88 · 2 tools 0.72 · 3 tools 0.62. Model requests: 220 answered, median 3.3 s; 109 backup requests and 221 retried transient errors kept the free tier inside each recording's window.

The 25 exact-match misses, read from the traces

KIND	ITEMS	WHAT HAPPENED
Format only	8	The right tool and value, worded differently (“the gym” for “Gym”)
Heard wrong	7	Spelled-out codes the recogniser misheard
Provider stalls	3	Google answered nothing for 28 s
Conditional	3	The expected answer contradicts the benchmark's mock data
Not passable	2	The expected value is never spoken
Judgement	2	DUET asked a question instead of acting

Where DUET sits among voice agents

SYSTEM	WHAT IT IS	WHEN THE USER CORRECTS THEMSELVES	FDB-V3 TOOL USE
GPT-Realtime	One speech-to-speech model	Acts on partial words: self-corrections 0.59	`Pass@1 0.600`
Gemini Live 3.1	One speech-to-speech model	Pre-emptive calls lock in stale values: 0.35	`Pass@1 0.540`
NVIDIA frontend–backend (2026)	A fast frontend hands tools to a backend	The closest design to ours, with no gate on actions	`Pass@1 0.44–0.48`
Moshi / Unmute, MiniCPM-o	Open full-duplex models	Interruptible and often fast; acting safely is not addressed	not reported
LiveKit Agents, Pipecat	Voice-agent frameworks	Barge-in, but no commit gate or ledger	—
Hume EVI	A voice that reads prosody	Tone shapes the reply, not the actions	—
DUET	A cascade with two minds and a coordinator	Epochs, commit gate, ledger, and tone that may ask before acting	`Pass@1 0.81 · 75/100 exact`

Baselines from the FDB-v3 paper and the projects' own publications. DUET's 0.81 was judged by Gemma 4 standing in for gpt-4o.